Homework 09
Use only when explicitly invoked to fit and compare SVM classifiers for binary or multiclass tasks.

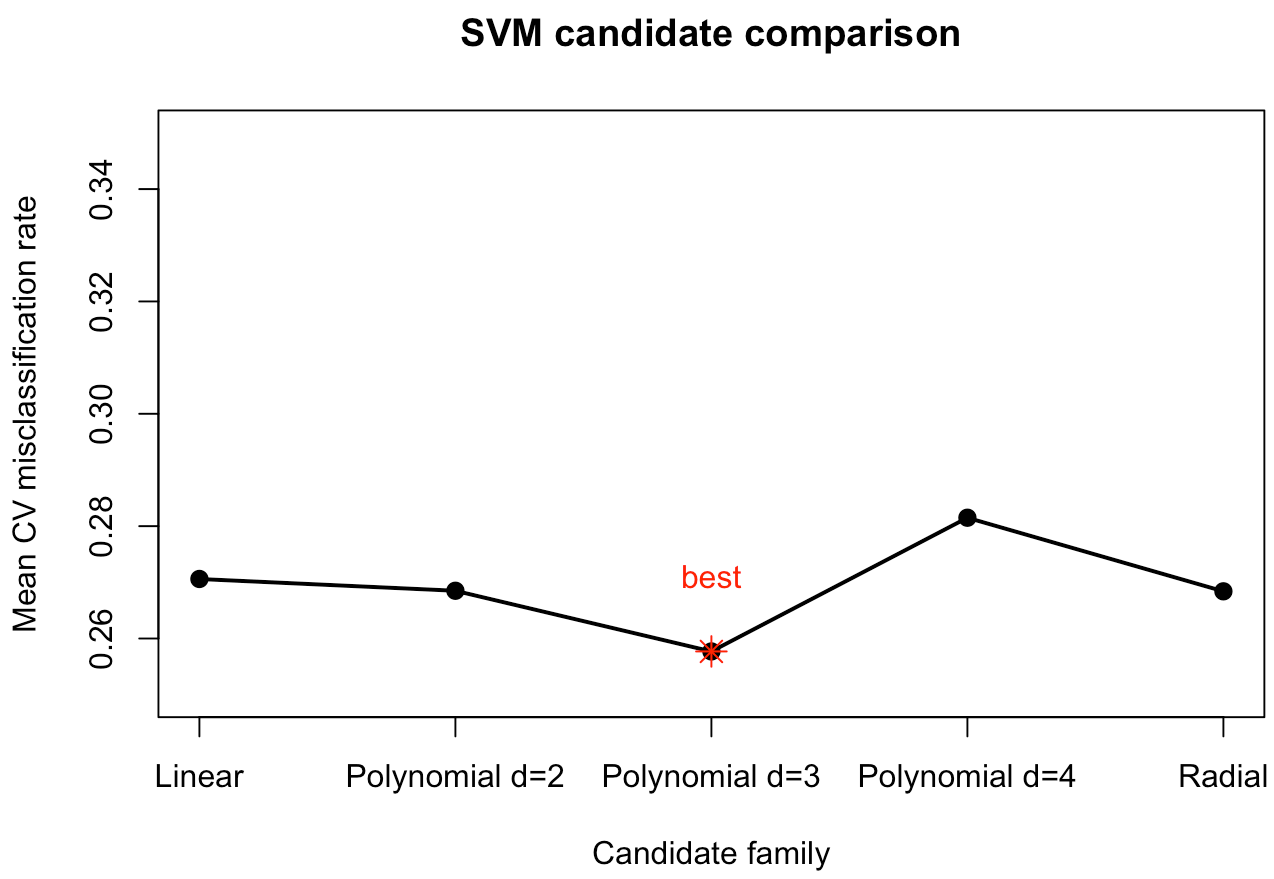
SVM analysis report

I explicitly invoked the `svm-analysis-skill` to analyze the SAheart data, using the response `chd`, the same nine predictors, and `famhist` encoded as `Absent = 1` and `Present = 2`. The skill directed the workflow: check suitability, choose kernel families, tune within the training data only, use the same five folds for all candidates, standardize within each fold, and select the model with the lowest mean cross-validation misclassification rate.

Candidate summary

The comparison used 19 settings total: linear SVMs with four values of C; polynomial kernels with degrees 2, 3, and 4 at costs 0.1, 1, and 10; and radial kernels with $C \in \{0.1, 1, 10\}$ and $\gamma \in \{0.01, 0.1, 1.0\}$. I held the polynomial kernel constant at `coef0 = 1` and used the same five folds across all candidates.

Kernel	Settings	Mean CV error
Linear	$C = 0.1, 1, 10, 100$	0.2706
Polynomial	degree 2, 3, 4; $C = 0.1, 1, 10$	0.2577 (best: degree 3, $C = 0.1$)
Radial	$C = 0.1, 1, 10$; $\gamma = 0.01, 0.1, 1.0$	0.2684 (best: $C = 10$, $\gamma = 0.01$)



Mean CV misclassification rates for candidate SVMs

Selected model and interpretation

The selected model is a polynomial SVM with degree 3 and cost $C = 0.1$. Its mean five-fold cross-validation misclassification rate was 0.2577, which was the best among the 19 settings considered. The radial family was close behind but slightly worse, and the linear kernel was easiest to interpret but less flexible.

This indicates that a moderate polynomial boundary captured the signal better than a purely linear rule or the tested radial settings for this data. The result should be read as the best model among this predefined SVM grid under the stated validation scheme, not as proof that no other kernel or parameter combination could work better.

Limitations

The comparison is restricted to the candidate families and tuning grid above. Misclassification rate treats the two classes equally and does not separate sensitivity and specificity. The selected settings depend on the fold splits and the search grid, so the ranking provides evidence for this specific study design but not a universal conclusion.

Full `SKILL.md` contents

```
---
name: svm-analysis-skill
description: Use only when explicitly invoked to fit and compare SVM classifiers for binary or multiclass
tasks.
disable-model-invocation: true
---

# SVM analysis

Use the dataset, response, and predictors supplied by the user. Check whether the task is appropriate for
SVMs, and explain any limitation or stop condition. Use only SVM-based models and do not add unrelated
methods.

Consider a small, justified set of kernels and parameters, such as linear, polynomial, and radial basis
kernels, and explain why each candidate is included. Choose tuning parameters and a tuning strategy inside
the training data only. Use the same folds for all candidates, and estimate any preprocessing from the
training folds before applying it to validation folds.

Select the final model by the lowest mean validation error, with misclassification rate as the default
summary. Report the model family, tuning settings, candidate table, and validation error. Then fit the
selected model on all training data and summarize the key result in a concise, evidence-based way.

Produce a compact one-page report with one figure and one table, 11-point font, and a brief interpretation
and limitation statement. Keep the workflow independent of any dataset-specific names or observed results.
```